MD RAKIBUL ISLAM RAIHAN

ML Systems & LLM Inference
PhD Applicant · Available July 2027 · Open to Relocation · Xi'an, China
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EDUCATION

M.Eng. in Software Engineering, School of Software, Northwestern Polytechnical University, Xi'an, China
Sep 2024 - Jul 2027 (expected) · Advisor: Prof. Zhang Peng
GPA: 88/100 (Top 1%); Chinese Government Scholarship (2024-2027)

B.Eng. in Computer Science & Technology, School of Computer Science, Northwestern Polytechnical University, Xi'an, China
Sep 2020 - Jul 2024 · Advisor: Dr. Yaxing Chen
GPA: 85/100 (Top 1%); Presidential Scholarship; Best Undergraduate Thesis Award

RESEARCH INTERESTS

My research centers on efficient machine learning systems, with an emphasis on low-precision quantization for generative models and large-scale distributed LLM inference. I work on FP8/FP4/NVFP4 quantization and quantization-aware training, custom Triton/CUDA kernels, multimodal audio-visual generative models and deepfake detection, and KV-cache-centric serving architectures that trade storage for computation. A second line of work concerns privacy-preserving distributed data systems built on homomorphic encryption.

PUBLICATIONS

First Author — MD Rakibul Islam Raihan, "NVFP4-DiT: Efficient 4-Bit Audio-Guided Video Diffusion Transformers for Low-Cost Video Generation" — IEEE Transactions on Neural Networks and Learning Systems (TNNLS, Q1), manuscript under review.

Theses — M.Eng.: "Detecting Deepfake Video by a Multimodal Audio-Visual Framework with Temporal Inconsistencies." B.Eng.: "Design and Implementation of a Distributed Confidential Query Protocol for Apache Spark."

RESEARCH EXPERIENCE

Graduate Researcher, Audio, Speech and Language Processing Lab (ASLP@NPU), School of Computer Science, Northwestern Polytechnical University, Sep 2024 - Present
Advisor: Prof. Zhang Peng

- Developed **NVFP4-DiT**, a **4-bit audio-guided video diffusion transformer**, with custom FP4-packed **Triton** kernels that are **3.2× faster than FP16** (2.1× over naive FP4) and cut memory **4×** at competitive generation quality.
- Implemented adaptive block-wise **FP8/FP4/NVFP4** quantization and quantization-aware training with learnable per-head cross-modal scales; evaluated trade-offs on WebVid-10M, VGGSound, and UCF-101.
- Built a **vLLM-style serving scheduler** (frame-level paged attention, dynamic frame batching) and benchmarked on **H100 GPUs** using PyTorch Profiler and Nsight to guide low-precision kernel choices.
- Research on multimodal deepfake detection using visual, audio, and temporal consistency analysis.
- M.Eng. Thesis: "Detecting Deepfake Video by a Multimodal Audio-Visual Framework with Temporal Inconsistencies."

Undergraduate Researcher, School of Computer Science, Northwestern Polytechnical University, Jan 2023 - Jul 2024
Advisor: Dr. Yaxing Chen (now Deputy Director, NWPU Human-Cyber-Physical Fusion Intelligent Computing Innovation Center)

- Designed and built a distributed confidential-query framework on Apache Spark + CKKS homomorphic encryption, validated on a **10-node cluster**: **95% privacy preservation** and **20% performance improvement** via optimized analytics pipelines.
- B.Eng. Thesis: "Design and Implementation of a Distributed Confidential Query Protocol for Apache Spark."

PROJECTS

NVFP4-DiT

A 4-bit audio-guided video diffusion transformer built with custom FP4-packed Triton kernels that are 3.2× faster than FP16 (2.1× over naive FP4) and cut memory 4× at competitive generation quality. Adds adaptive block-wise FP8/FP4/NVFP4 quantization-aware training with learnable per-head cross-modal scales, evaluated on WebVid-10M, VGGSound and UCF-101, plus a vLLM-style serving scheduler with frame-level paged attention benchmarked on H100 GPUs.

AI Infrastructure Portfolio — github.com/theraihanrakibb/ai-infra-portfolio

Ten Dockerized AI-infrastructure services - LLM gateway, model serving, GPU observability, KV-cache simulation, benchmarking, quantization, tracing and RAG - each with CI and a reproducible quickstart.

Distributed Training Launcher — github.com/theraihanrakibb/train-launcher

A PyTorch distributed CLI that launches and supervises multi-node training jobs with fault detection, automatic retry and checkpoint recovery.

INDUSTRY EXPERIENCE

AI Infrastructure Engineer Intern, InfiX.ai, Shenzhen, China · Apr 2026 - Jun 2026

Production LLM serving on 4× A800-80GB and multi-node H200: 3-tier KV-cache offload over Mooncake RDMA (8.3× lower long-context TTFT, 55s→6.7s), 2+2 prefill/decode disaggregation (192→256 concurrent requests), and changes to SGLang cache-management and memory-allocation internals.

Software Engineer Intern (AI Agent), Hong Kong Amraim Group Co., Ltd., Shenzhen, China · Jun 2025 - Sep 2025

Delivered two end-to-end full-stack applications (React + RESTful services) and AI agents that automated marketing workflows, lifting team productivity by 25%.

Electrical Software Engineer Intern, Shaanxi Longong Intelligent Technology Co., Ltd., Xi'an, China · Feb 2025 - Apr 2025

Real-time CODESYS PLC motor-control software over Modbus/CAN/Ethernet-IP; 15% faster system response to IEC 61131 industrial safety standards.

AWARDS AND HONORS

- Chinese Government Scholarship, 2024-2027
- Presidential Scholarship, Northwestern Polytechnical University, 2020-2024
- Wu Yajun Scholarship, 2024
- Best Undergraduate Thesis Award, 2024
- Outstanding Graduate, Northwestern Polytechnical University, 2024
- Hackathon Winner, NWPU AI Challenge, 2023

SERVICES & LEADERSHIP

- President, International Student Union (ISU), Northwestern Polytechnical University

TECHNICAL SKILLS

LLM Inference & Systems: SGLang, vLLM, KV Cache, PD Disaggregation, Mooncake/RDMA, Continuous Batching, RadixAttention

GPU & Performance: CUDA, CUDA C++, Triton, Nsight Systems, Nsight Compute, PyTorch Profiler, GEMM optimization, FP8/FP4/INT4

ML & Multimodal: PyTorch, Hugging Face Transformers, Diffusion/DiT, Multimodal Learning, OpenCV

Distributed Systems: PyTorch Distributed, Docker, Kubernetes, Linux, Prometheus, CI/CD

Programming: Python, C++, CUDA C++, Bash, SQL, Data Structures & Algorithms

CERTIFICATIONS

- **Deep Learning Specialization**, DeepLearning.AI.
- **Claude Academy: Claude Code in Action**, Anthropic, issued Oct 2026. Credential ID d9a951d15c796f242db8113c0d796c3d.
- **AI Agents Fundamentals**, Hugging Face, issued Oct 2026. Credential ID theraihanrakib.